



**Database System Final Project
(HOSPITAL MANAGEMENT SYSTEM)**

(BS-IT-F23 Morning)

INSTRUCTOR:

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1. Working:

The Hospital Management System automates the administrative and clinical operations of a hospital. It manages patient information, appointments, billing, staff payroll, inventory, and other hospital functions. The system improves operational efficiency, reduces administrative workload, and enhances patient care by streamlining processes. It ensures accurate data management and facilitates quick decision-making by providing real-time insights through various queries and reports.



2. Entities and Attributes:

1. Patient

- **Primary key:** PatientID
- **Attributes:** Name, Age, Gender, Address, Phone

2. Doctors

- **Primary key:** DoctorID
- **Attributes:** Name, Speciality, Contact

3. Department

- **Primary key:** DeptID
- **Attributes:** Name

4. Appointment

- **Primary key:** AppointmentID
- **Attributes:** Date, Time, Purpose, PatientID (Foreign Key), DoctorID (Foreign Key), BillNo

5. Bills

- **Primary key:** BillNo
- **Attributes:** Amount, Date, PaymentMode

6. Staff

- **Primary key:** StaffID
- **Attributes:** Name, Role, DeptID(FK), Contact

7. Payroll

- **Primary key:** PayID
- **Attributes:** Salary, PayDate, StaffID (Foreign Key)

8. Administration

- **Primary key:** Name
- **Attributes:** Contact

9. Inventory

- **Primary key:** ItemID
- **Attributes:** ItemName, Quantity, Category, DepID (Foreign Key)

Relationships:

1. Appointment and Bills

- **Associative Entity**

2. Patient and Appointment

- **Associative Entity**

3. Appointment and Doctors

- **Associative Entity**

4. Doctors and Department

- **Relationship:** Works

5. Administration and Inventory

- **Relationship:** Manages

6. Staff and Payroll

- **Relationship:** Receives

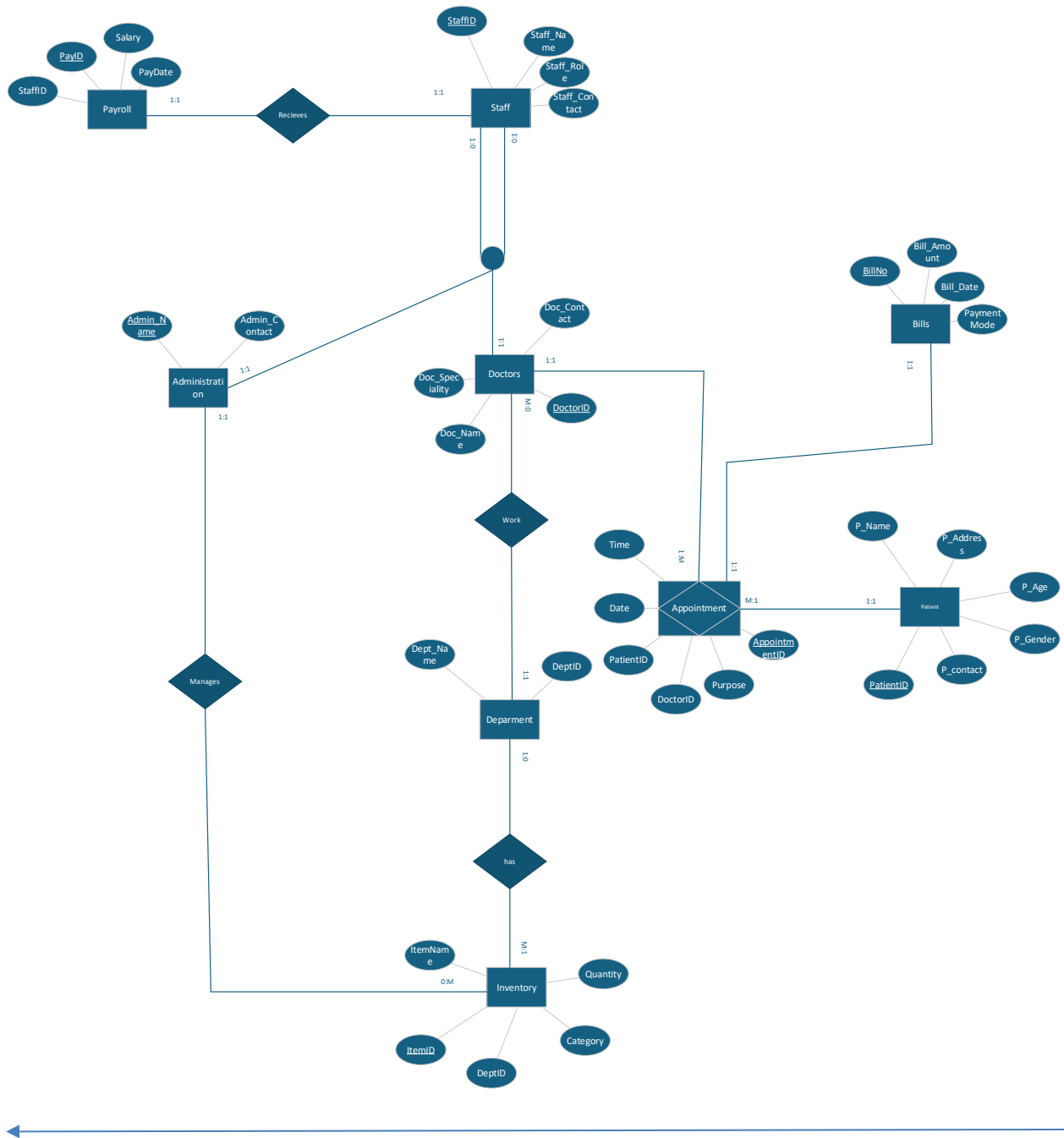
7. Administration and Staff

- **Relationship:** Manages

8. Inventory and Department

- **Relationship:** has

ERD



3. RELATIONAL SCHEMA

- Top down approach

1. Patient (PatientID (PK), P_Name, P_Age, P_Gender, P_Address, P_Contact)
2. Inventory(ItemID (PK), ItemName, Quantity, Category)
3. Bills(BillNo (PK), Bill_Amount, Bill_Date, PaymentMode)
4. Staff(StaffID (PK), Staff_Name, Staff_Role, Staff_Contact, DeptID(FK))
5. Payroll(PayID (PK), Salary, PayDate, StaffID (FK))
6. Doctors(StaffID(PK,FK) , Doc_Speciality)
7. Appointments(AppointmentID (PK), AppointmentDate, Time, Purpose, PatientID (FK), StaffID (FK))
8. Department(DeptID (PK), Dept_Name)

- **Bottom up approach**

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PatientID , P_Name, P_Age, P_Gender, P_Address,
P_Contact, ItemID , ItemName, Quantity, Category, BillNo,
Bill_Amount, Bill_Date, PaymentMode, StaffID , Staff_Name,
Staff_Role, Staff_Contact, PayID , Salary, PayDate, Doc_Speciality,
AppointmentID , AppointmentDate, Time, Purpose, DeptID ,
Dept_Name

)

➤ **1NF :**

- Patient(PatientID, P_Name, P_Age, P_Gender, P_Address, P_Contact)
- Inventory(ItemID, ItemName, Quantity, Category)

- Bills(BillNo, Bill_Amount, Bill_Date, PaymentMode, PatientID)
- Staff(StaffID, Staff_Name, Staff_Role, Staff_Contact, DeptID)
- Payroll(PayID, Salary, PayDate, StaffID)
- Doctors(StaffID, Doc_Speciality)
- Appointments(AppointmentID, AppointmentDate, Time, Purpose, PatientID, StaffID)
- Department(DeptID, Dept_Name)

➤ **2NF:**

THESE ARE ALREADY IN 2NF BECAUSE THERE ARE NO PARTIAL FUNCTIONAL DEPENDENCIES.

➤ **3NF:**

THESE ARE ALREADY IN 3NF BECAUSE THERE ARE NO TRANSITIVE DEPENDENCIES.

4. **DESCRIPTION:**

Patient

Attribute	Data Type	Size	Constraints
PatientID	Number	-	Primary key
P_Name	Varchar2	15	Not Null
P_Gender	Char	1	Check(P_Gender IN('M','F')
P_Age	Number	-	-
P_Address	Varchar2	50	NOT Null
P_contact	Varchar2	11	NOT Null

Inventory

Attribute	Data Type	Size	Constraints
itemID	Number		Primary Key
itemName	Varchar2	15	Not Null
Quantity	Number		
Category	Varchar2	15	Not Null

Bills

Attribute	Data Type	Size	Constraints
BillNo	Number		Primary Key
Bill_Amount	Number		
Bill_Date	Date		Default
PaymentMode	Varchar2	15	
PatientID	Number		Foreign Key(PateintID)

Staff

Attribute	Data Type	Size	Constraints
DeptID	Number		Foreign Key(DeptID)
StaffID	Number		PrimaryKey
Staff_name	Varchar2	15	Not Null
Staff_Role	Varchar2	15	Not Null
Staff_contact	Varchar2	11	Not Null

Payroll

Attribute	Data Type	Size	Constraints
PayID	Number		Primary key
Salary	Number		
PayDate	Date		Default
StaffID	Number		Foreign key(StaffID)

Doctors

Attribute	Data Type	Size	Constraints
StaffID	Number		Primary key ,Foreign key
Doctor_speciality	Varchar2	20	Not Null

Appointment

Attribute	Data Type	Size	Constraints
AppointmentID	Number		Primary Key
AppointmentDate	Date		Default
Time	Varchar2	7	
Purpose	Varchar2	15	Not Null
PatientID	Number		Foreign Key(PatientID)
StaffID	Number		Foreign Key(StaffID)

Department

Attribute	Data Type	Size	Constraints
DeptID	Number		Primary key
Dept_Name	Varchar2	15	Not Null



5. CREATE-Statements:

```
CREATE TABLE Patient (  
    PatientID NUMBER CONSTRAINT pk_patientid PRIMARY KEY,  
    P_Name VARCHAR2(15) NOT NULL,  
    P_Gender CHAR(1) CONSTRAINT c_p_gender CHECK (P_Gender IN ('M', 'F')) ,  
    P_Age NUMBER,  
    P_Address VARCHAR2(50) NOT NULL,  
    P_contact VARCHAR2(11) NOT NULL  
);
```

```
CREATE TABLE Inventory (  
    ItemID NUMBER CONSTRAINT pk_itemid PRIMARY KEY,  
    ItemName VARCHAR2(15) NOT NULL,  
    Quantity NUMBER,  
    Category VARCHAR2(15) NOT NULL  
);
```

```
CREATE TABLE Bills (  
    BillNo NUMBER CONSTRAINT pk_billno PRIMARY KEY,  
    Bill_Amount NUMBER,  
    Bill_Date DATE DEFAULT SYSDATE,  
    PaymentMode VARCHAR2(15),  
    PatientID NUMBER,  
    CONSTRAINT fk_bills_patient FOREIGN KEY (PatientID) REFERENCES Patient (PatientID)  
);
```

```
CREATE TABLE Department (  
    DeptID NUMBER CONSTRAINT pk_deptid PRIMARY KEY,  
    Dept_Name VARCHAR2(15) NOT NULL  
);
```

```
CREATE TABLE Staff (  
    StaffID NUMBER CONSTRAINT pk_staffid PRIMARY KEY,
```

```
Staff_Name VARCHAR2(15) NOT NULL,  
Staff_Role VARCHAR2(15) NOT NULL,  
Staff_contact VARCHAR2(11) NOT NULL,  
DeptID CONSTRAINT fk_staff_department REFERENCES Department (DeptID)  
);
```

```
CREATE TABLE Payroll (  
    PayID NUMBER CONSTRAINT pk_payid PRIMARY KEY,  
    Salary NUMBER,  
    PayDate DATE DEFAULT SYSDATE,  
    StaffID NUMBER,  
    CONSTRAINT fk_payroll_staff FOREIGN KEY (StaffID) REFERENCES Staff (StaffID)  
);
```

```
CREATE TABLE Doctors (  
    StaffID NUMBER CONSTRAINT pk_staffid_doctor PRIMARY KEY,  
    Doctor_speciality VARCHAR2(20) NOT NULL,  
    CONSTRAINT fk_doctors_staff FOREIGN KEY (StaffID) REFERENCES Staff (StaffID)  
);
```

```
CREATE TABLE Appointment (  
    AppointmentID NUMBER CONSTRAINT pk_appointmentid PRIMARY KEY,  
    AppointmentDate DATE DEFAULT SYSDATE,  
    Time VARCHAR2(7),
```

```
Purpose VARCHAR2(15) NOT NULL,  
  
PatientID NUMBER,  
  
StaffID NUMBER,  
  
CONSTRAINT fk_appointment_patient FOREIGN KEY (PatientID) REFERENCES Patient  
(PatientID),  
  
CONSTRAINT fk_appointment_staff FOREIGN KEY (StaffID) REFERENCES Staff (StaffID)  
  
);
```



6. VIEWS:

1. DoctorAppointmentsView:

- IT PROVIDES AN OVERVIEW OF APPOINTMENTS WITH DOCTOR DETAILS AND PATIENT NAMES.

```
CREATE OR REPLACE VIEW DoctorAppointmentsView AS
```

```
SELECT A.AppointmentID, A.AppointmentDate, A.Time, A.Purpose, P.P_Name AS  
PatientName, S.Staff_Name AS DoctorName, D.Doctor_speciality
```

```
FROM Appointment A
```

```
JOIN Patient P
```

```
ON A.PatientID = P.PatientID
```

```
JOIN Staff S
```

```
ON A.StaffID = S.StaffID
```

```
JOIN Doctors D
```

```
ON S.StaffID = D.StaffID;
```

2. DepartmentStaffView:

- IT PROVIDES ALL STAFF MEMBERS ALONG WITH THEIR DEPT NAMES.

```
CREATE OR REPLACE VIEW DepartmentStaffView AS
```

```
SELECT S.StaffID, S.Staff_Name, S.Staff_Role, D.Dept_Name AS Department
FROM Staff S
JOIN Department D
ON S.DeptID = D.DeptID;
```

7. SELECT STATEMENTS:

1. Appointments for a Specific Date:

```
SELECT A.AppointmentID, A.AppointmentDate, A.Time, A.Purpose, P.P_Name AS PatientName,
S.Staff_Name AS DoctorName
FROM Appointment A
JOIN Patient P
ON A.PatientID = P.PatientID
JOIN Staff S
ON A.StaffID = S.StaffID
WHERE A.AppointmentDate = (select AppointmentDate from appointment where
appointmentid=1 )
```

2. Doctors and Their Specialties:

```
SELECT S.StaffID, S.Staff_Name, D.Doctor_speciality
FROM Doctors D
JOIN Staff S
ON D.StaffID = S.StaffID;
```

3. Total Salary by Department:

```
SELECT DeptID, SUM(Salary) AS TotalSalary
FROM Payroll P
JOIN Staff S
ON P.StaffID = S.StaffID
GROUP BY DeptID;
```

4. Patients Treated by a Specific Doctor:

```
SELECT P.PatientID, P.P_Name, A.AppointmentDate, A.Purpose
FROM Appointment A
JOIN Patient P
ON A.PatientID = P.PatientID
WHERE A.StaffID = 101;
```

5. Staff Contact Details by Role:

```
SELECT S.Staff_Name, S.Staff_Role, S.Staff_contact  
FROM Staff S  
WHERE S.Staff_Role LIKE '%Nurse%';
```



MODIFICATIONS:

1. ADDED DEPARTMENT ENTITY SO THAT RADIOLOGY AND PATHOLOGY ARE HANDELED WITHIN IT.
2. ADDED PHYSICIAN ENTITY TO HANDLE PATIENT AND PHYSICIAN DIRECTLY
3. ADDED APPOINTMENT ENTITY TO MAKE A DIRECT LINK BETWEEN PATIENT AND PHYSICIAN.
4. ADDED INVENTORY ENTITY TO COUNT ROOMS AND BEDS FOR PATIENTS.